

Epidemic modelling for children

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Abstract

The dynamics of epidemics are poorly understood with, typically, the general population parroting the exponential growth model of Malthus that was debunked by Darwin. Statistical modelling by Farr had already shown that was not the case for epidemics, the subsequent ordinary differential equations of the standard SIR model of Kermack & McKendrick made that explicit. There are stochastic extensions to that latter model, there are also other stochastic models with different average reproduction rates which haven't supplanted the standard model. All these models, quantitative and qualitative, require somewhat more Mathematical ability than that required by the model of Malthus. Here we show that a game of tag, an agent-based model, averages to the standard model corrected to meet the definition of the basic reproduction number by using basic probability theory. This model is easily explainable to children and can also be viewed as a game of cards. It also supplants numerous stochastic models that cannot match its reproduction number. Apart from being stochastic it is trivially extensible to animal social networks where a comparison can be made to flattening the curve. The use of sparse graphs also allows for modelling the geographic spread of an epidemic.

Keywords: Epidemiological Model, Agent-Based Modelling, ABM, Stochastic, SIR, exponential growth, selection without replacement

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1. Introduction

Do you know how an epidemic grows? If you believe it is exponentially then can you explain what happens and when to cause a finite variable to leave a curve to infinity? Exponential growth is acknowledged to be a poorly understood concept and Occam's Razor suggests that another growth rate is applicable. On social media it is possible to converse with people who won't concede an argument because they have their own definition of exponential, they actually meant approximate, etc. etc. Science relies on common definitions for communication so, for the avoidance of doubt, exponential growth occurs when the independent variable of the exponential function is increased by constant steps. One particular example is an increasing geometric series leading to the alternative term of geometric growth. Note that the exponential function of a non-linear function cannot grow exponentially.

There was no intent in the previous paragraph to patronise the reader. Establishing this context was deemed to be important because it is possible to cite the work of numerous public health officials, popular science authors, academics, etc., etc. contained in books, articles, papers, videos and official briefings that epidemics grow exponentially. However, it doesn't seem fair to name any of those people without a comprehensive review and to compile such a large review does not seem to be fair on the author. Thus, it is left to the reader to trust their own memories or to perform their own search.

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That exponential growth is so poorly understood is strange given its long history. Muir [1] found that the Sumerians were calculating compound interest over 4000 years ago. The economist Malthus [2] proposed that populations grow exponentially whilst resources grow linearly. The public health statistician Farr [3] fitted both the Normal and truncated Cubic Exponential distributions to infection data. For this work of not drawing a straight line on a log plot posterity has named a Law after Farr. In his seminal work Darwin [4] explicitly called out Malthus and debunked his idea over the space of two paragraphs using primary school level Mathematics and a *Reductio ad absurdum* argument.¹ In what has subsequently become known as the standard SIR model Kermack and McKendrick [7] described a system of three ordinary differential equations in which the infection curve arises due to competition between logistic growth and exponential decay.

The standard SIR model is an example of a compartmental model, as introduced by Ross and Hudson. A constant population of size N is split into the compartments S (susceptible to infection), I (infected and infectious) and R (recovered and immune to re-infection).

$$N = S + I + R \quad (1)$$

Differentiating equation 1 with respect to time, t , and re-arranging results in an equation where the reproduction number, $-dS/dR$, has been replaced without loss of generality by the product of its maximum value, the basic reproduction number R_0 , and some proportion X .

$$\frac{dI}{dt} = \left(-\frac{dS}{dR} - 1 \right) \frac{dR}{dt} \equiv (R_0 X - 1) \frac{dR}{dt}, \quad X \in [0, 1] \quad (2)$$

Kermack and McKendrick closed the model by assuming that a cycle of infection takes place over a constant time interval of $1/\gamma$ resulting in exponential decay leading to X now being the ratio of growth to exponential growth. The growth itself being modelled as proportional to the product of the number of infected and susceptible which results in logistic growth.

$$\frac{dI}{dt} = \left(R_0 \frac{S}{N} - 1 \right) \gamma I \quad (3)$$

Dropping the variable R but retaining the closure of Kermack and McKendrick produces the SI model which has the analytical solution of the logistic function. This function is merely a scaled and shifted version of the hyperbolic tangent as typically found on scientific calculators. It is a sigmoid, or S-shaped, curve that has approximately exponential growth from zero at its start and approximately exponential decay as it increases to one at its end. Hence describing epidemic growth as approximately exponential when, for a specific case, it is actually logistic causes issues with deciding how approximate is approximate and when it is no longer approximate given that the error will be growing exponentially.

$$N = S + I, \quad \frac{dI}{dt} = R_0 \frac{S}{N} \gamma I \Rightarrow \frac{dI}{dt} = R_0 \frac{N - I}{N} \gamma I \Rightarrow \frac{I}{N} = \frac{1}{1 + \exp(-R_0 \gamma t)} = \frac{1 + \tanh(R_0 \gamma t/2)}{2} \quad (4)$$

The Kermack-McKendrick model has the adjective standard because it fits a number of datasets. Other growth rates are available, some in modifications to the standard SIR model, others in stochastic models such as the chain binomial models which began with the qualitative model of Reed-Frost [8]. This is a somewhat confusing state of affairs given that these models are typically applied to the same disease outbreaks when only one can be true and thus match the data. Given the plethora of models, it is easy to assume that they outnumber the available datasets.

Teaching the standard SIR model to the general population is not possible without first teaching them calculus. Is there no alternative to the simplistic exponential growth model that causes so much confusion, even to those who should know better? Unfortunately, that would require yet another model, preferably one that matched the data.

¹Darwin was ignored by Lotka[5] and Volterra[6] in their model of predators and prey, a mainstay of Mathematical Biology textbooks.

2. An agent-based model on a complete graph

Mathematics modelling concerns itself with abstraction thus an epidemic can be thought of as a child's game of tag with just two rules:

1. When first tagged, tag two (R_0) others at random.
2. Ignore any subsequent tag.

Note that there are three states that a player can be in - never tagged, tagging and stopped tagging. These map directly to the compartments of the standard SIR model hence it can be considered to be the minimal SIR model. For the disease to have exponential growth to the end the never tagged have to be mercilessly hunted down whilst all others are ignored. i.e. The disease has to have a memory.

For the sake of reproducibility, the children were replaced by a computer code. The implementation required a split between the random selection and the state of a player. This made for a card game in which each child is replaced by two copies of the same card. To simulate the tagging one pack is shuffled and two cards are dealt, if one of those corresponds to the tagger then a third card is dealt. In the code this is achieved with a Fisher-Yates [9] shuffle using a pseudo-random number generator and a set seed. The other pack is used to track the compartments by creating a set of columns corresponding to which generation a card is in when it is first dealt. Thus, the first column has one card, the second two cards and the third has between zero and four cards and so on. Handily the columns form an epidemic curve. Using a population equal to some power of two minus one, such as 31, makes it obvious when comparing the outcome of a game to exponential growth.

How realistic is this model? The game of tag uses *selection without replacement* which means that the act of tagging is equivalent to sampling R_0 times from a Hypergeometric distribution which has S chances of success in a population of size $N - 1$. This needs to be applied repeatedly for all infected I in a generation, combining the distributions is the same as sampling $R_0 I$ times with S chances of success. Thus, the difference equation for the average behaviour using the expected value of the distribution is

$$I_{n+1} - I_n = \left(R_0 \frac{S_n}{N - 1} - 1 \right) I_n \quad (5)$$

This is equivalent to equation 3 discretised over the cycle $1/\gamma$ and corrected to meet the definition of the basic reproduction number. i.e. $I_n = 1, S_n = N - 1 \Rightarrow I_{n+1} = R_0$. Games of 1000 individuals were played 100 times and the results averaged rather than the much simpler method of just evaluating the distribution. Four different values of R_0 were used - 2, 3, 4 and 5 - with the calculated growth rates being seen to match the formula. It should be noted that *selection with replacement* results in the Binomial distribution with the same expected value for each tagger and allows for an individual tagged multiple times over a cycle to be treated as successes.

The odds of the game matching exponential growth may be calculated in a similar manner to the solution of the birthday problem. For 31 individuals the probability decreases from certainty for the first three individuals, to about evens after eight individuals before shrinking to about 2 in a trillion for the last individual.

$$\frac{31}{31} \times \frac{30}{30} \times \frac{29}{29} \times \frac{28}{30} \times \frac{27}{29} \times \cdots \times \frac{3}{29} \times \frac{2}{30} \times \frac{1}{29} = \frac{31!}{31 \times 30^{15} \times 29^{15}} \quad (6)$$

3. Extension to social networks

Agent-based models are usually heterogeneous however the game is, with the exception of chance, homogeneous. The standard SIR model is said to apply to well mixed populations, in the case of the game the population forms a complete graph. It is easily extended to a sparse graph by reducing the choice of each agent. This was done for an animal social network of 1610 nodes obtained as the largest subgraph from a study of the mixing of voles across their geographical territories by Davis et al. [10] in which the degree of the nodes varied from 1 to 37. The raw data being acquired from the archive of Rossi and Ahmed [11]. Each and every node had patient zero assigned to it and the calculations were repeated 10 times with R_0 equal to 2.

The average results compared to a complete graph of the same size can be seen in figure 1. These results may be interpreted as the complete graph representing normal behaviour and the social network representing a change in

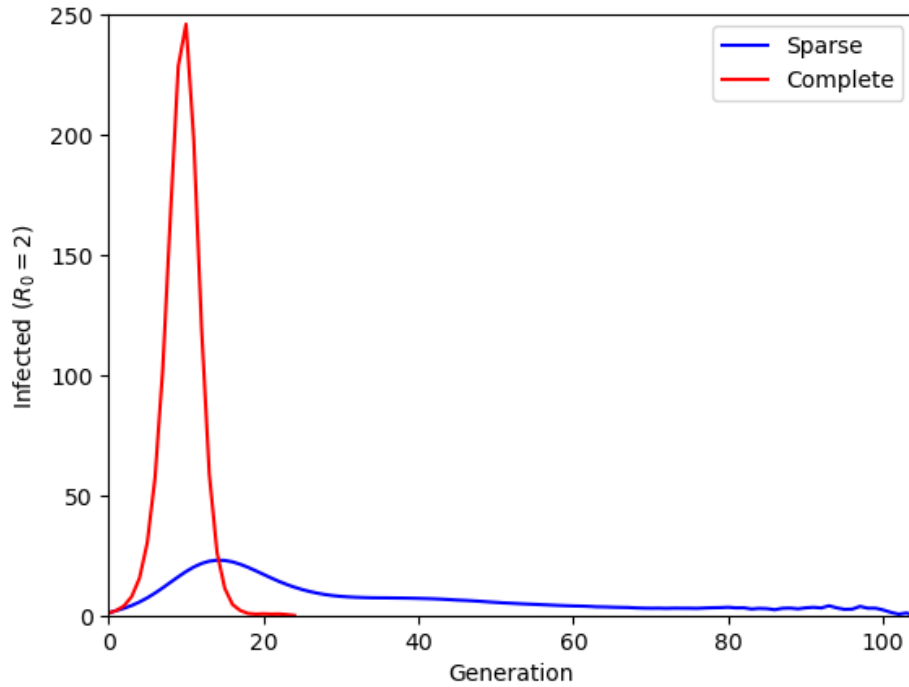


Figure 1. Flattening of the curve due to a sparse graph

Plain additive noise	σ is constant
Scaled additive noise	$\sigma = \sqrt{\gamma R_0 IS/N}$
External parameter noise	$\sigma \propto \gamma R_0 IS/N$
Heterogeneous parameter noise	$\sigma \propto \gamma R_0 \sqrt{IS/N}$

Table 1. Textbook noise

behaviour to flatten the curve. The epidemic on the social network took 105 generations to complete leaving 892 voles never being infected whereas the complete graph lasted only 25 generations and left 323 individuals with 'herd immunity'. Hence, in the absence of other measures, flattening the curve drags out an epidemic and results in further waves due to a lack of acquired immunity.

4. Discussion

The minimal SIR model supplants numerous others. Perhaps it will make the exponential growth model of the Malthusians finally disappear. The chain binomial models, like the minimal SIR model, are discrete stochastic models. They have an average form that has not supplanted the standard SIR model thus the minimal SIR model should supplant them. The textbook by Keeling and Rohani [12] details four Wiener process extensions to the standard SIR model with the different standard deviations given in table 1. A Hypergeometric distribution with a small number of samples can be approximated by a Binomial distribution which will scale to the Normal distribution used by a Wiener process. There being four items in the table rather than one is the first hint as to their applicability and the other is the variance of the Minimal SIR model as given by equation 7.

$$\sigma^2 = R_0 I \cdot \frac{S}{N-1} \cdot \frac{N-1-S}{N-1} \cdot \frac{N-1-R_0 I}{N-2} \quad (7)$$

The two games are accessible to primary school children and illustrate that Mathematics is all about abstraction. It is trivial to add prior immunity to both games, additionally the card game can be easily altered to exponential growth

by never returning a dealt card to the pack. To aid with education simple JavaScript implementations of both games have been made as well as NetLogo variants of the SI, SIR and SIS models. Some of the Mathematical analysis may be fine for pre-university students based on whether they study probability theory or calculus. Their critical thinking may be affected by the realisation that pseudo-science that had been debunked long ago was being promoted during the Covid-19 pandemic and that a knowledge of Mathematics helps to detect it.

Do you know how an epidemic grows? (By selection without replacement. i.e. Sampling from a Hypergeometric distribution.) One infects two, two infect four or three or two or one or none. All with different probabilities and a weighted average of logistic growth.

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